Sampling Synthesis

Intro to Audio Tech 1
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Monday, September 22

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Lesson Outline

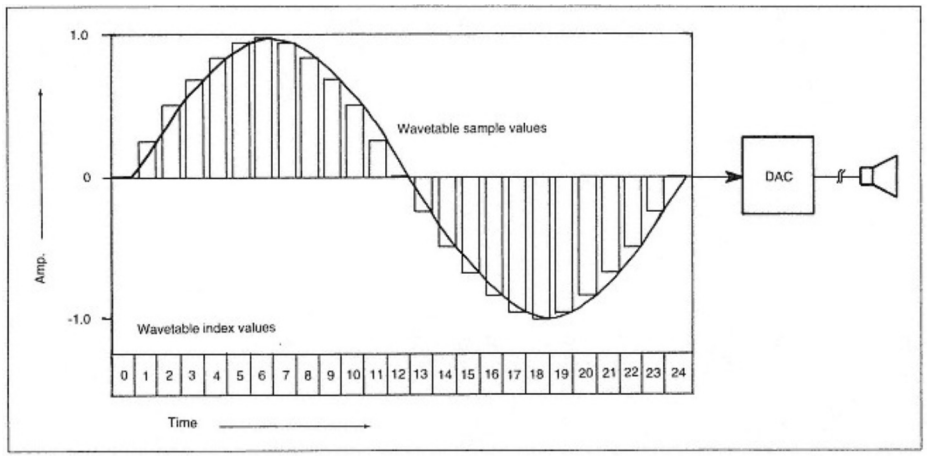
- Wavetable Refresher
- Sampling Synthesis Techniques
- HW2 Work Time
- Up Next…

Checkpoint 4

Wavetable (and Sampling) Synthesis

Wavetable

- Store waveform (full cycle) in a buffer/table
- Function to read values from the wavetable
  - The read point location is analogous to phase in waveform
- What would correspond to frequency?
  - Length of table and rate of increment



Sampling Synthesis

- Like a wavetable but uses prerecorded sounds
  - Can be seconds of a sound vs a wavetable is a single cycle
- Unlike other synthesis techniques we've discussed you start with the whole sound and can manipulate it after
- A sampler captures sounds and let's you organize, edit, and play them back

Sampling Synthesis - Techniques

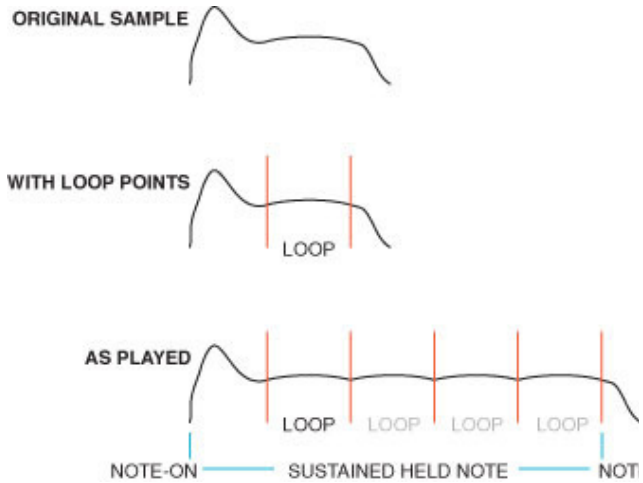
- Sampling synthesis is different from all other synthesis techniques because the audio content has already been defined
  - The length, frequency/amplitude relationships, etc
  - We cannot *fully* go in and adjust those parameters
- This can cause some issues for us that we use different techniques to work around

Chipmunking and Multi-Sampling

- Suppose we play a recording of a piano playing A4 (440 Hz)
- How would we change that sample to play A5 (880 Hz)?
  - Play it back at 2x speed
  - Will that sound like what A5 naturally sounds like on the piano?
  - What about if we play back at A6?
- When we speed up our playback we get a pitch shift, but this is over a shorter period with an equal shift to upper harmonics causing the “chipmunk effect”
- To combat this, we would use multisampling
  - We record many, many samples of our desired sound at different pitches and dynamics and store them in banks of similar timbres

Looping

- In wavetable synthesis, we continuously read across a wavetable to create a periodic waveform allowing us to control the length of the sound
- In sampling synthesis, we already have an inherit amplitude envelope and a complex waveform, so we can't just play the whole recording over and over to lengthen it
- We try to define loop points within a recording to play through while sustaining a "note" in the recording
- We need to be strategic with this and often still will hear the cyclical timbral pattern, but it can be helped by crossfading between loop points



Other terms and techniques related to samplers

- Cue time / skiptime and cue points:
  - Starting the recording playback in a different place than the start and applying a new amplitude envelope
- Reverse:
  - Playing the recorded samples backwards
- Stereo Panning:
  - E.g., Panning the low notes of a piano left and the highs of a piano right
- Slicing:
  - Cutting up a recording into new segments and rearranging their order e.g., Beat slicing

Working with Recorded Sounds in Max

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Objects for Recorded Sound Playback

- Wave~
  - Same as wavetable synthesis, playback from a buffer, loop points as arguments
  - Line~ or phasor from 0-1

- Groove~
  - Variable playback from a buffer, loop points as arguments
  - Takes signal in to set speed, takes message with start point as play, can also be looped

- Play~
  - Position-based playback, uses position and time as starts/stops

- Sfplay~
  - Play an audio file from disk (file on computer)

- Info~
  - Tells us info about our buffer

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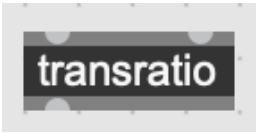
Playback of Recorded Sounds in Buffers

- In wavetable, we save one cycle, so if it is 1 sec long, we read through it as many times as we need our frequency in a second
- With recorded audio, it is already complex with many cycles and usually longer
  - It was already recorded and sampled where 1 second of audio = 44.1k
  - So normal speed playback is 44100 samples per second
- We need to read across the buffer for the duration of the recorded file
- With wave~
  - A line~ from 0-1 over the duration of the file
  - Take 1/the duration of the file to convert the seconds of playback to Hz for a phasor~

Pitch-shifting playback

- To change the pitch, we read through the file faster or slower
- Double is an octave up and half is an octave below
- But what if we wanted to do steps in between?
  - The ratio for a semitone is 2 ^ (semitones / 12)
- Max has two built in objects for this transposition
  - Transratio
  - Rtt.scala



Homework 3

Up Next…

- Homework 2 due Friday
- Exam 1 on Wednesday
- Readings